

A Sharp Bound for Minimal Dependencies among Powers of Linear Forms

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Abstract

We prove a sharp bound for minimal linear dependencies among powers of linear forms. Let $\ell_1, \dots, \ell_t$ be nonzero homogeneous linear forms in finitely many variables over a field of characteristic zero, and suppose that $\ell_1^m, \dots, \ell_t^m$ are linearly dependent while no proper subfamily of these powers is linearly dependent. If $L = \text{span}\{\ell_1, \dots, \ell_t\}$, then

$$\dim L \leq \frac{t + m - 2}{m}.$$

The bound is attained for infinitely many pairs $(t, \dim L)$, and hence both the leading constant and the displayed additive term are best possible along an infinite sequence of parameters. This answers a question of Bukh.

1 Introduction

Throughout the paper, a linear form means a homogeneous polynomial of degree one. Thus, after choosing coordinates $x_1, \dots, x_n$ on a vector space over a field K , a linear form has the shape

$$a_1x_1 + \dots + a_nx_n$$

with coefficients $a_i \in K$ and with no constant term. Equivalently, it is a linear functional on the underlying vector space, that is, an element of the dual vector space. We write $[t] = \{1, \dots, t\}$.

Bukh asked the following question. Suppose that $\ell_1(x), \dots, \ell_t(x)$ are linear forms whose m -th powers are linearly dependent, and no proper subset of these powers has this property. If $L = \text{span}\{\ell_1, \dots, \ell_t\}$, how large can $\dim L$ be? Bukh noted on his problem list that a bound of the form $\dim L \leq c_mt + O(1)$ is known with $c_m = 3/(m+4)$ for $m \geq 2$, and asked whether the same should hold with $c_m = 1/m$ [1]. Minimal dependencies among powers of linear forms also appear as an auxiliary ingredient in Bukh's work on extremal graphs without exponentially-small bicliques [2].

We prove the conjectured leading constant, with a sharp additive term for infinitely many values of t .

Theorem 1 (Main theorem). *Let K be a field of characteristic zero. Let $m \geq 1$, and let $\ell_1, \dots, \ell_t$ be nonzero homogeneous linear forms in finitely many variables over K . Suppose that $\ell_1^m, \dots, \ell_t^m$ are linearly dependent, but every proper subfamily of the displayed family of m -th powers is linearly independent. Put $L = \text{span}\{\ell_1, \dots, \ell_t\}$. Then*

$$\dim L \leq \frac{t + m - 2}{m}.$$

Equivalently, $\dim L \leq t/m + O_m(1)$.

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For $m = 1$, the statement reduces to the elementary fact that a circuit of t vectors has rank $t - 1$.

We now show, over any field K of characteristic zero, that the leading constant $1/m$ cannot be improved, and that the additive term in Theorem 1 is attained for infinitely many parameter values.

Fix $r \geq 1$. If $r = 1$, the equality example is the two-form family $\ell_1 = \ell_2 = x_0$. Thus assume $r \geq 2$ and put $M = m(r - 1)$ and $t = M + 2 = m(r - 1) + 2$. Choose pairwise distinct scalars $a_1, \dots, a_t \in K$. This is possible because every field of characteristic zero is infinite. In variables $x_0, \dots, x_{r-1}$, define

$$\ell_i = x_0 + a_i x_1 + a_i^2 x_2 + \dots + a_i^{r-1} x_{r-1}.$$

Then $\text{span}\{\ell_1, \dots, \ell_t\}$ has dimension r , because the coefficient matrix is Vandermonde of rank r .

Now consider ℓ_i^m . The coefficient of $x_0^{\alpha_0} x_1^{\alpha_1} \dots x_{r-1}^{\alpha_{r-1}}$ in ℓ_i^m is

$$\binom{m}{\alpha_0, \dots, \alpha_{r-1}} a_i^{\alpha_1 + 2\alpha_2 + \dots + (r-1)\alpha_{r-1}}.$$

Thus each coefficient of ℓ_i^m , as a function of a_i , is a scalar multiple of a monomial a_i^s with $0 \leq s \leq m(r - 1) = M$. Hence the coefficient row space of $\ell_1^m, \dots, \ell_t^m$ is contained in the $(M + 1)$ -dimensional space spanned by

$$(1, \dots, 1), \quad (a_i)_{i=1}^t, \quad (a_i^2)_{i=1}^t, \quad \dots, \quad (a_i^M)_{i=1}^t.$$

Since $t = M + 2$, the t forms $\ell_1^m, \dots, \ell_t^m$ are linearly dependent.

It remains to check minimality. For every integer $0 \leq s \leq M$, one can write $s = j_1 + \dots + j_m$ with $0 \leq j_\nu \leq r - 1$. Indeed, write $s = q(r - 1) + b$ with $0 \leq b < r - 1$. If $b = 0$, use q summands equal to $r - 1$ and the remaining summands equal to 0. If $b > 0$, then $q \leq m - 1$ because $s \leq m(r - 1)$; use q summands equal to $r - 1$, one summand equal to b , and the remaining summands equal to 0. Taking $\alpha_j = \#\{\nu : j_\nu = j\}$ gives a monomial of total degree m whose coefficient row is a nonzero scalar multiple of $(a_i^s)_{i=1}^t$. This scalar is nonzero because $\text{char } K = 0$. Therefore the coefficient matrix contains the full Vandermonde block

$$(a_i^s)_{0 \leq s \leq M, 1 \leq i \leq t}.$$

Thus any choice of $M + 1 = t - 1$ columns gives a square Vandermonde matrix with distinct parameters a_i , and hence is nonsingular. Hence any proper subfamily of size $t - 1$ among $\ell_1^m, \dots, \ell_t^m$ is linearly independent. A fortiori every smaller proper subfamily is linearly independent.

Thus we have constructed a minimally dependent family with $\dim L = r$ and $t = m(r - 1) + 2$. Consequently, $r = (t + m - 2)/m$, so equality holds in Theorem 1 for infinitely many values of t .

Since equality holds for $t = m(r - 1) + 2$, no leading constant smaller than $1/m$ can hold in general.

Corollary 2. *The optimal leading constant in Bukh's question is $c_m = 1/m$.*

Remark 3 (Veronese-circuit form). *Equivalently, let W be a finite-dimensional vector space over a field K of characteristic zero, and let*

$$\nu_m : \mathbb{P}(W) \rightarrow \mathbb{P}(\text{Sym}^m W), \quad [\ell] \mapsto [\ell^m],$$

be the m -th Veronese embedding. If $p_1, \dots, p_t$ form a circuit after applying ν_m , and if $d = \dim \langle p_1, \dots, p_t \rangle$, then $t \geq md + 2$. Indeed, $d + 1 = \dim \text{span}\{\ell_1, \dots, \ell_t\}$, so this is exactly Theorem 1. The rational normal curve construction above shows that the bound is sharp.

Corollary 4. *Let K have characteristic zero, and let $X \subseteq \mathbb{P}^n(K)$ be a finite set. Suppose that every projective flat Π of dimension d satisfies $|X \cap \Pi| \leq md + 1$. Then the degree- m Veronese images of the points of X are linearly independent.*

Proof. If the Veronese images were dependent, choose a minimally dependent subfamily. If its original points span a projective d -flat, the preceding remark gives at least $md + 2$ points in that flat, contradicting the hypothesis. $\square$

Remark 5 (Diagonal circuits). *Over a field of characteristic zero, if*

$$\sum_{i \in [t]} c_i \ell_i^m \equiv 0$$

is a support-minimal diagonal identity, then $\dim \text{span}\{\ell_1, \dots, \ell_t\} \leq (t+m-2)/m$. Thus every minimal zero $\Sigma \wedge \Sigma$ circuit of top fan-in t and degree m has rank at most $(t+m-2)/m$.

Remark 6 (Characteristic). *The proof also works over any infinite field of characteristic $p > m$, since the relevant multinomial coefficients do not vanish. Over finite fields one may base-change to the algebraic closure. The restriction on the characteristic is necessary for this argument: when $p \leq m$, Frobenius phenomena may cause the Veronese map to degenerate.*

The proof translates the problem into the language of Schur powers of linear codes. The only additive-combinatorial input is the following Kneser-type inequality for coordinatewise products of vector spaces. It is the trivial-stabilizer case of the Schur-product Kneser theorem of Mirandola and Zémor [3, Theorem 3.3]; we recall a proof in Section 2 to make the note self-contained.

2 A Schur-product Kneser inequality

Throughout this section, K is an infinite field. For vectors in K^t , multiplication is coordinatewise. If $S, T \subseteq K^t$ are linear subspaces, their product is $ST = \text{span}\{s * t : s \in S, t \in T\}$. For a subspace $V \subseteq K^t$, define $\text{St}(V) = \{u \in K^t : uV \subseteq V\}$. The scalar line $K\mathbf{1}$, where $\mathbf{1} = (1, \dots, 1)$, is always contained in $\text{St}(V)$. We say that $V \subseteq K^t$ has *full support* if, for every coordinate i , some vector in V has nonzero i -th coordinate. A vector $x \in K^t$ is *invertible* if all its coordinates are nonzero. Let $V^\times$ denote the set of invertible vectors in V .

Lemma 7. *If $V \subseteq K^t$ has full support, then V has a basis consisting of invertible vectors.*

Proof. Choose a basis $v_1, \dots, v_k$ of V . For $\alpha \in K$, put

$$y_\alpha = v_1 + \alpha v_2 + \dots + \alpha^{k-1} v_k.$$

For every coordinate j , the j -th coordinate of y_α is a polynomial in α . Since V has full support, this polynomial is not identically zero. The bad values of α are therefore contained in the union of finitely many proper zero sets of nonzero univariate polynomials, hence form a finite set. Because K is infinite, there are infinitely many α for which y_α is invertible. Choosing k distinct such values of α gives a basis, since the change-of-basis matrix is Vandermonde. $\square$

Lemma 8. *There are only finitely many unital subalgebras of K^t under coordinatewise multiplication.*

Proof. Let $H \subseteq K^t$ be a unital subalgebra. Define an equivalence relation on $\{1, \dots, t\}$ by

$$i \sim j \iff h_i = h_j \text{ for every } h \in H.$$

Then H is contained in the space of vectors that are constant on the blocks of this partition.

Conversely, fix a block B of the partition. For each other block B' , by definition of the partition there is some $h_{B'} \in H$ whose value on B differs from its value on B' . Here $h(B)$ denotes the common value of h on the block B . The vector

$$\prod_{B' \neq B} \frac{h_{B'} - h_{B'}(B')\mathbf{1}}{h_{B'}(B) - h_{B'}(B')}$$

belongs to H , is equal to 1 on B , and is equal to 0 on every other block. Thus H is exactly the algebra of vectors constant on the blocks of its partition. Since there are only finitely many partitions of $\{1, \dots, t\}$, there are only finitely many such algebras. $\square$

Lemma 9 (e-transform lemma). *Let $S, T \subseteq K^t$ be nonzero subspaces. Suppose that T has a basis of invertible vectors. For every $x \in S^\times$ there exist a unital subalgebra $H_x \subseteq K^t$ and a subspace $V_x \subseteq K^t$ such that*

$$H_x V_x = V_x, \quad xT \subseteq V_x \subseteq ST,$$

and

$$\dim V_x + \dim H_x \geq \dim S + \dim T.$$

Proof. First reduce to the case $x = \mathbf{1} \in S$. Put $S' = x^{-1}S$. If the case $\mathbf{1} \in S'$ gives a unital subalgebra H and a subspace V' with

$$HV' = V', \quad T \subseteq V' \subseteq S'T,$$

and

$$\dim V' + \dim H \geq \dim S + \dim T,$$

then $V_x = xV'$ satisfies

$$HV_x = V_x, \quad xT \subseteq V_x \subseteq ST, \quad \dim V_x + \dim H \geq \dim S + \dim T.$$

Thus it suffices to prove the case $x = \mathbf{1} \in S$.

We prove this case by induction on $k = \dim S$. If $k = 1$, then $S = K\mathbf{1}$. Take $H_x = K\mathbf{1}$ and $V_x = T$.

Assume $k > 1$. For an invertible vector $e \in T$, define $S(e) = S \cap Te^{-1}$ and $T(e) = T + Se$. Since $\mathbf{1} \in S$ and $e \in T$, we have $\mathbf{1} \in S(e)$, so $S(e) \neq 0$. To see the dimension formula, consider the linear map

$$\phi_e: S \rightarrow K^t/T, \quad s \mapsto se + T.$$

Then $\ker \phi_e = S \cap Te^{-1} = S(e)$, and $T(e)/T \simeq \text{im } \phi_e$. Hence

$$\dim T(e) = \dim T + \dim S - \dim S(e),$$

or equivalently $\dim S(e) + \dim T(e) = \dim S + \dim T$. Also $S(e)T(e) \subseteq ST$. Indeed, $S(e)T \subseteq ST$, and if $a \in S(e)$, then $a = be^{-1}$ for some $b \in T$, so for $s \in S$ we have $a(se) = bs \in ST$.

If $S(e) = S$ for every $e \in T^\times$, then $S \subseteq Te^{-1}$, and therefore $Se \subseteq T$ for every invertible $e \in T$. Since the invertible vectors in T span T , it follows that $ST \subseteq T$. As $\mathbf{1} \in S$, also $T \subseteq ST$, hence $ST = T$. Let H be the unital subalgebra generated by S , and let $V = T$. Since $ST = T$, repeated multiplication by elements of S preserves T , and hence $HV = V$. Also $T \subseteq V \subseteq ST$, and

$$\dim V + \dim H \geq \dim T + \dim S.$$

Otherwise, choose $e \in T^\times$ with $0 < \dim S(e) < \dim S$. Since T has a basis of invertible vectors, T has full support. As $T \subseteq T(e)$, the space $T(e)$ also has full support, and by Lemma 7 it has a basis of invertible vectors. Applying the induction hypothesis to the pair $S(e), T(e)$ yields a unital subalgebra H and a subspace V with

$$HV = V, \quad T \subseteq T(e) \subseteq V \subseteq S(e)T(e) \subseteq ST,$$

and

$$\dim V + \dim H \geq \dim S(e) + \dim T(e) = \dim S + \dim T.$$

This completes the induction. □

Theorem 10 (Schur-product Kneser, trivial-stabilizer case). *Let K be an infinite field. Let $S, T \subseteq K^t$ be nonzero subspaces. If $\text{St}(ST) = K\mathbf{1}$, then $\dim ST \geq \dim S + \dim T - 1$.*

Proof. Since $\text{St}(ST) = K\mathbf{1}$, the product ST has full support; otherwise arbitrary values on the unused coordinates would give a larger stabilizer. Hence both S and T have full support. By Lemma 7, T has a basis of invertible vectors, and S has infinitely many invertible vectors.

Let $k = \dim S$. Choose a basis $s_1, \dots, s_k$ of S and put

$$y_\alpha = s_1 + \alpha s_2 + \dots + \alpha^{k-1} s_k.$$

As in Lemma 7, there are infinitely many α such that y_α is invertible. For each such y_α , Lemma 9 gives a unital subalgebra H_{y_α} and a space V_{y_α} .

By Lemma 8, there are only finitely many possible subalgebras. Hence there exist k distinct scalars $\alpha_1, \dots, \alpha_k$ such that $y_{\alpha_1}, \dots, y_{\alpha_k}$ are invertible, form a basis of S , and have the same associated algebra H . Write $x_i = y_{\alpha_i}$ and $V_i = V_{x_i}$.

Since $x_1, \dots, x_k$ is a basis of S , we have $ST = x_1T + \dots + x_kT$. As $x_iT \subseteq V_i \subseteq ST$ for every i , it follows that $ST = V_1 + \dots + V_k$. Since each V_i is H -stable and $ST = V_1 + \dots + V_k$, the space ST is H -stable. Hence $H \subseteq \text{St}(ST) = K\mathbf{1}$. Thus $\dim H = 1$. For any i , Lemma 9 gives

$$\dim ST + 1 \geq \dim V_i + \dim H \geq \dim S + \dim T.$$

Therefore

$$\dim ST \geq \dim S + \dim T - 1.$$

□

3 Proof of the main theorem

We first translate the problem into Schur powers. Let $r = \dim L$. Choose a basis $x_1, \dots, x_r$ for L . Write

$$\ell_i = a_{1i}x_1 + \dots + a_{ri}x_r, \quad i = 1, \dots, t.$$

Let $A = (a_{ji})$ be the corresponding $r \times t$ matrix, and let $C \subseteq K^t$ be the row space of A . Then $\dim C = r$.

For a subspace $C \subseteq K^t$, define its Schur powers by

$$C^{\langle 1 \rangle} = C, \quad C^{\langle j+1 \rangle} = CC^{\langle j \rangle}.$$

We shall use the elementary identity $C^{\langle a \rangle}C^{\langle b \rangle} = C^{\langle a+b \rangle}$, which follows directly from associativity and commutativity of coordinatewise multiplication.

Lemma 11. *For every $j \geq 1$, the rank of the family $\ell_1^j, \dots, \ell_t^j$ equals $\dim C^{\langle j \rangle}$.*

Proof. The rank of the family $\ell_1^j, \dots, \ell_t^j$ equals the dimension of the row space of their coefficient matrix in the monomial basis of homogeneous degree- j polynomials in $x_1, \dots, x_r$. Indeed, since $x_1, \dots, x_r$ are linearly independent linear forms, they may be extended to a coordinate system on the ambient dual space. In particular, the degree- j monomials in $x_1, \dots, x_r$ are linearly independent.

Expand

$$\ell_i^j = (a_{1i}x_1 + \dots + a_{ri}x_r)^j.$$

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_r)$ with $|\alpha| = j$, the coefficient of $x_1^{\alpha_1} \dots x_r^{\alpha_r}$, as i varies, is

$$\binom{j}{\alpha} (a_{1i}^{\alpha_1} \dots a_{ri}^{\alpha_r})_{i=1}^t,$$

that is, a nonzero scalar multiple of a coordinatewise product of j rows of A . Conversely, every product of j basis rows of C appears in this way, up to a nonzero multinomial factor. Since every element of C is a linear combination of the rows of A , the coordinatewise product of any j elements of C is, by multilinearity, a linear combination of coordinatewise products of j basis rows of A . Since $\text{char } K = 0$, these multinomial factors do not vanish. Hence the coefficient row space of $\ell_1^j, \dots, \ell_t^j$ is precisely $C^{\langle j \rangle}$. □

Because $\ell_1^m, \dots, \ell_t^m$ are minimally linearly dependent, their rank is $t - 1$. By Lemma 11, $\dim C^{\langle m \rangle} = t - 1$. Moreover, the linear relation is unique up to scalars and has full support: if a relation had some zero coefficient, then the remaining proper subfamily of $\ell_1^m, \dots, \ell_t^m$ would already be linearly dependent, contradicting minimality. Thus there exist nonzero scalars $\lambda_1, \dots, \lambda_t \in K^\times$ such that

$$\lambda_1 \ell_1^m + \dots + \lambda_t \ell_t^m = 0.$$

The vector $\lambda = (\lambda_1, \dots, \lambda_t)$ annihilates the coefficient row space of $\ell_1^m, \dots, \ell_t^m$, which is $C^{\langle m \rangle}$. Thus

$$C^{\langle m \rangle} = \left\{ z = (z_1, \dots, z_t) \in K^t : \sum_{i \in [t]} \lambda_i z_i = 0 \right\}.$$

So $C^{\langle m \rangle}$ is a hyperplane whose normal vector has no zero coordinate. Since every $\lambda_i \neq 0$, for each coordinate j one can choose z in this hyperplane with $z_j \neq 0$; hence it has full support.

We next record the stabilizer property needed to apply Theorem 10.

Lemma 12. *Let*

$$H = \left\{ z \in K^t : \sum_{i \in [t]} \lambda_i z_i = 0 \right\},$$

where each $\lambda_i \neq 0$. Then $\text{St}(H) = K\mathbf{1}$.

Proof. Let $u = (u_1, \dots, u_t) \in \text{St}(H)$. Then for every $z \in H$, $\sum_{i \in [t]} \lambda_i u_i z_i = 0$. Thus the linear functional $z \mapsto \sum_{i \in [t]} \lambda_i u_i z_i$ vanishes on the hyperplane $H = \ker(z \mapsto \sum_{i \in [t]} \lambda_i z_i)$. Since both linear functionals vanish on the same codimension-one subspace H , they are scalar multiples of each other. Hence there exists $c \in K$ such that $\lambda_i u_i = c \lambda_i$ for all i . Since $\lambda_i \neq 0$, we obtain $u_i = c$ for all i . Hence $u \in K\mathbf{1}$. $\square$

Lemma 13. *For every $1 \leq j \leq m$, $\text{St}(C^{\langle j \rangle}) = K\mathbf{1}$.*

Proof. For $j = m$, this follows from Lemma 12, because $C^{\langle m \rangle}$ is the hyperplane $\sum_{i \in [t]} \lambda_i z_i = 0$ with all $\lambda_i \neq 0$.

Now let $1 \leq j < m$, and suppose that $u \in \text{St}(C^{\langle j \rangle})$. Then $uC^{\langle j \rangle} \subseteq C^{\langle j \rangle}$. By the Schur-power identity,

$$C^{\langle j \rangle} C^{\langle m-j \rangle} = C^{\langle m \rangle}.$$

Therefore

$$uC^{\langle m \rangle} = uC^{\langle j \rangle} C^{\langle m-j \rangle} \subseteq C^{\langle j \rangle} C^{\langle m-j \rangle} = C^{\langle m \rangle}.$$

Therefore $u \in \text{St}(C^{\langle m \rangle}) = K\mathbf{1}$. Since $K\mathbf{1} \subseteq \text{St}(C^{\langle j \rangle})$ is automatic, equality follows. $\square$

We now finish the proof of Theorem 1. Let $d_j = \dim C^{\langle j \rangle}$. Then $d_1 = \dim C = r$. For $1 \leq j < m$, apply Theorem 10 to $S = C$ and $T = C^{\langle j \rangle}$. Since $ST = C^{\langle j+1 \rangle}$ and, by Lemma 13, $\text{St}(C^{\langle j+1 \rangle}) = K\mathbf{1}$, we have

$$d_{j+1} = \dim C^{\langle j+1 \rangle} \geq \dim C + \dim C^{\langle j \rangle} - 1 = r + d_j - 1.$$

Iterating this inequality from $d_1 = r$ gives $d_j \geq r + (j-1)(r-1) = jr - (j-1)$ for all $1 \leq j \leq m$. Taking $j = m$ gives $d_m \geq mr - (m-1)$. But minimal dependence gives $d_m = t-1$. Therefore $t-1 \geq mr - (m-1)$, or equivalently,

$$r \leq \frac{t+m-2}{m}.$$

Since $r = \dim L$, Theorem 1 follows.

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Declaration on the use of AI

The authors used generative AI tools to assist with discussing proof strategies, proof checking, and exposition. All mathematical arguments, results, and conclusions were reviewed and verified by the authors.

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